Time Series Modelling Case Study Assignment
Forecasting Electricity Demand

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GitHub: https://github.com/TEJENDRASAI/Advance-Data-Science-A-1.git

1.Preface:

According to the data, there is a clear yearly rhythm in German power demand, with summertime demand being lower and wintertime demand rising. Because a significant portion of the signal was already captured by just repeating the same week from the prior year, the seasonal naïve forecast became a challenging criterion to surpass. The study consequently became less about establishing that a complex model is always superior, and more about questioning which model contributes meaningful information beyond this yearly memory. Random Forest, Gradient boosting, and weekly aggregated LSTM were the top weekly models. SARIMA and SARIMAX were beneficial diagnostically but did not beat the seasonal naïve benchmark in this run. Random Forest is the best balanced choice for a workable weekly operational model because it outperforms the benchmark, is simpler to understand and maintain than LSTM, and mostly relies on lagged load, weather, holiday, and calendar data.

2. Time-Series Evidence and Data Preparation:

The data is taken from the publicly available data from https://data.open-power-system-data.org/time_series/. From there we have retrieved the data of only German country as per the assignment requirements and the German country is identifies as 'DE'. After we have kept the data from January 1at 2015 until the end and made some initial plots of Exploratory data analysis and made the data to weekly and daily values. And the temperature data is directly taken from https://archive-api.open-meteo.com/v1/archive

Weekly averaging was employed for the benchmark, SARIMA, SARIMAX and feature-based models because it keeps the annual signal evident while minimizing hourly noise. While the original hourly series was retained for the LSTM because neural sequence models can profit from the daily load cycle that vanishes after weekly aggregation, daily values were still helpful during exploration. Below Figure 1 is important because it shows why a yearly benchmark is not a weak baseline. Most years follow a similar broad shape: demand is relatively high in colder weeks, falls through the middle of the year, and rises again toward winter. The 2020 line is less regular, which matters because much of the test period overlaps with unusual social and economic conditions.

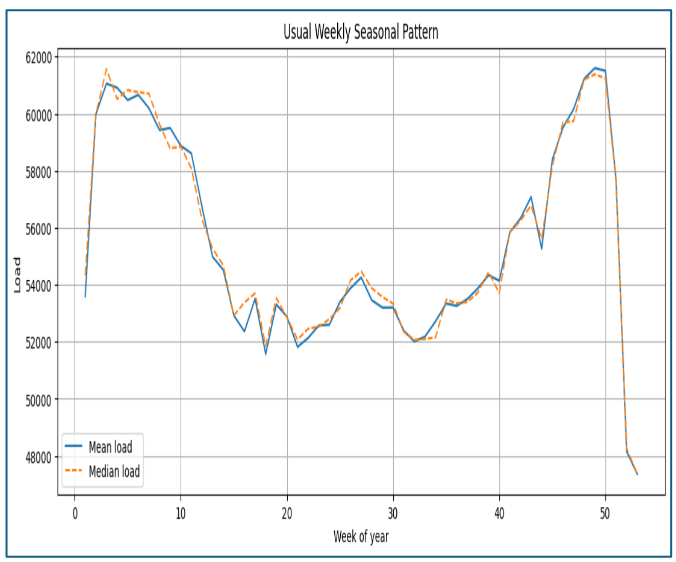


Figure 1. Usual weekly seasonal pattern in German electricity demand, showing recurring annual structure.

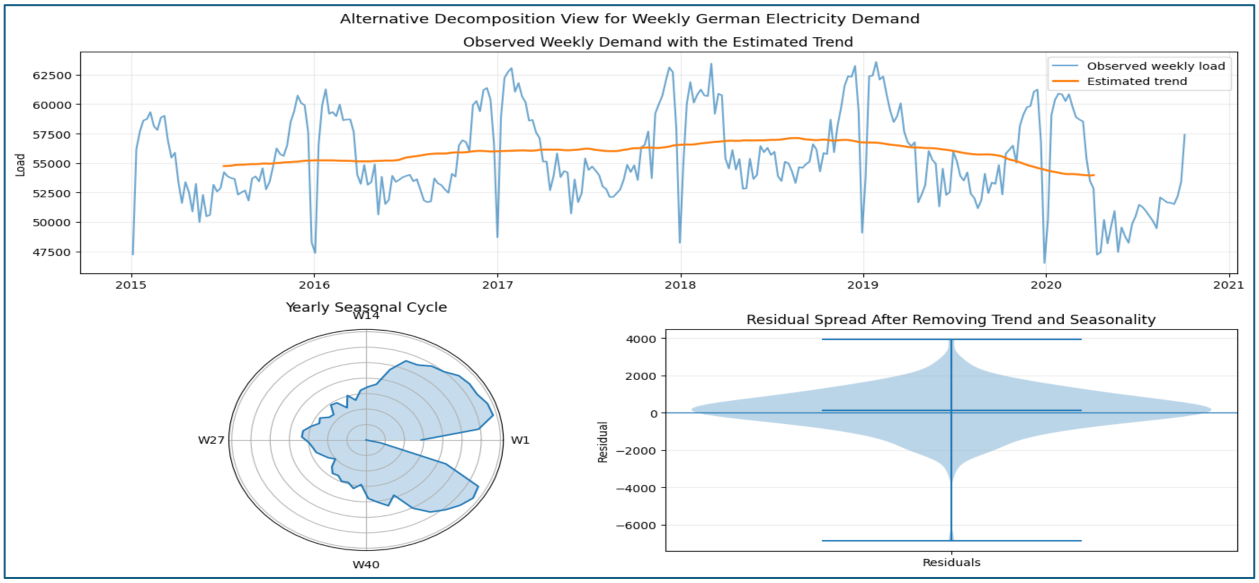


Figure 1.2.Decomposition View

Series	ADF stat.	ADF p	KPSS	KPSS p	Interpretation
Original weekly load	-4.047	0.00118	0.160	0.10	Statistically usable, but seasonality remains visible.
First difference	-7.069	<0.001	0.056	0.10	Stronger stationarity evidence after removing week-to-week level movement.
Seasonal difference	-4.295	0.00045	1.342	0.01	Seasonal change still has structure, so diagnostics must be interpreted carefully.

Table 1. Stationarity from ADF and KPSS.

Stationarity was checked using both ADF and KPSS tests because they frame the null hypothesis differently. The original weekly series had ADF $p = 0.00118$ and KPSS $p = 0.10$, so it was not a simple random walk. However, the ACF/PACF and seasonal decomposition still supported a seasonal period of 52 weeks. The conclusion is therefore balanced: the series can be modelled statistically, but its calendar structure should not be ignored.

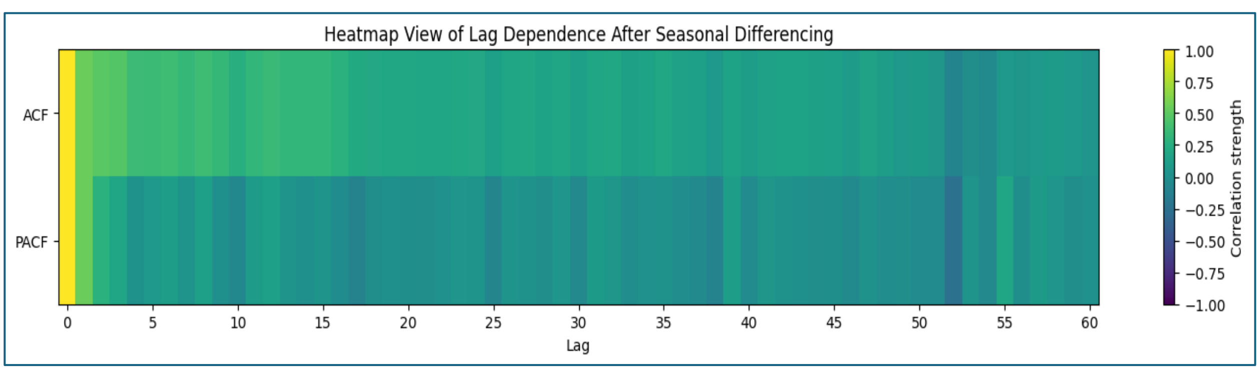


Figure 2. Lag-dependence heatmap after seasonal differencing, used to inspect remaining autocorrelation.

3.Models Built and Why They Were Used:

The process purposefully transitioned from basic to sophisticated models. Mean, naïve, drift and seasonal naive forecasts were fitted initially so that following models had to prove they provided value. The seasonal naïve model was the major reference since electricity consumption has a natural yearly cycle.

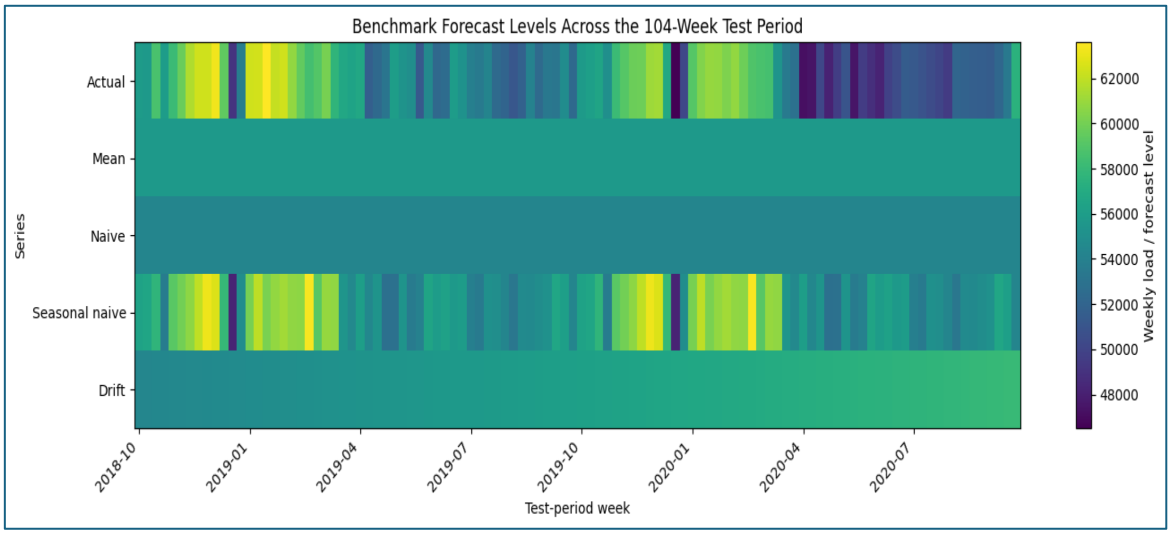


Figure 3. Benchmark forecast levels across the 104-week test period.

Model	Why it was included	What's the outcome
Seasonal naive	Checks whether last year's same week is already enough.	Best benchmark is RMSE 3006.76.
SARIMA	Tests whether classical autocorrelation and seasonal differencing explain demand.	Good diagnostic model, but poor hold-out accuracy.
SARIMAX	Adds weather and holidays to test conditional explanatory value.	Better than SARIMA but still weaker than benchmark.
Gradient Boosting	Learns non-linear effects from lagged load, weather and calendar variables.	Improves on seasonal naive; interpretable through importance.
Random Forest	Just to compare the both models so that we ca also know which models performs better on this data	Best balanced weekly operational choice after `lstm.
LSTM	Uses hourly sequences to learn daily and weekly shape directly.	Best numerical result after weekly aggregation.

Table 2. Purpose of each model and its current result interpretation.

Next, all non-seasonal orders p = 0 to 6, d = 0 to 2, and q = 0 to 6 were tried using a SARIMA grid search, and the most likely specification was chosen using AIC. In addition to meeting the assignment's parameter-search criteria, this reduced the subjectivity of the model selection. SARIMA (2,2,6) (1,1,1,52) was the chosen weekly model. To determine whether significant structure persisted after fitting, residual plots, residual ACF, and the forecast interval were examined. The SARIMA forecast was less accurate than the benchmark, which can be explained by the residual diagnostics, which indicated that some structure persisted after fitting.

SARIMAX was then used to add temperature and holiday variables. Berlin temperature was employed as a useful weather proxy and turned into heating and cooling degree characteristics. Public holidays can reduce the demand for commercial and industrial electricity, which is why they were included in the holiday count. Although the complete weather-and-holiday SARIMAX outperformed SARIMA, seasonal naive was still superior. This outcome is helpful because it demonstrates that unless the model structure can make good use of the variables, adding them does not ensure progress.

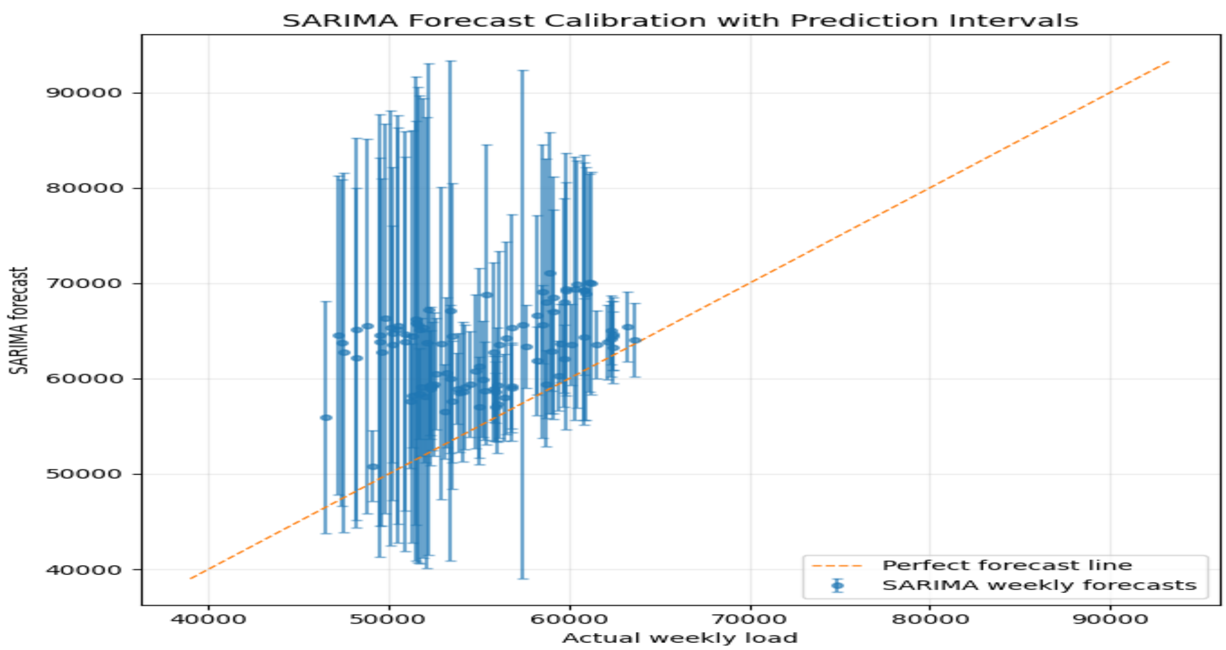
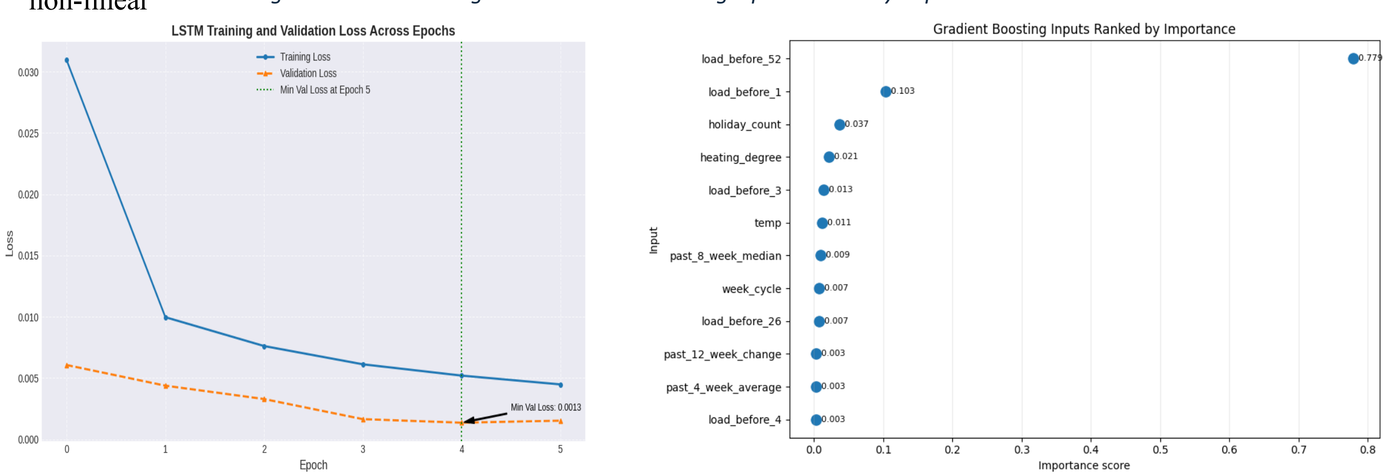


Figure 4. SARIMA forecast calibration with prediction intervals

4.LSTM and Feature Models:

Lagged demand, rolling summaries, temperature, degree days, holidays, and cyclical calendar variables were all included in the feature-based model set. Because machine-learning regressors may simulate non-linear interactions without requiring the weekly series to be stationary, this method was included Another feature-based ensemble model that was tested versus Gradient Boosting was Random Forest. This allowed two non-linear

Figure 5. LSTM Training loss & Gradient Boosting inputs ranked by importance



models with identical lagged load, weather, holiday, and calendar inputs to be compared. This model family's two-year residual plot, feature importance, and final cross-model error table provide forecast evidence.

To learn daily and intra-week trends that weekly aggregation eliminates, the LSTM model employed hourly data. Three lookback designs one-day, one-week, and two-week memory were examined, and the scaler was only attached during the training phase. The one-week memory LSTM was best among the neural settings. Although it is more challenging to maintain and explain, it delivered the lowest error after aggregating its hourly estimates down to weekly means.

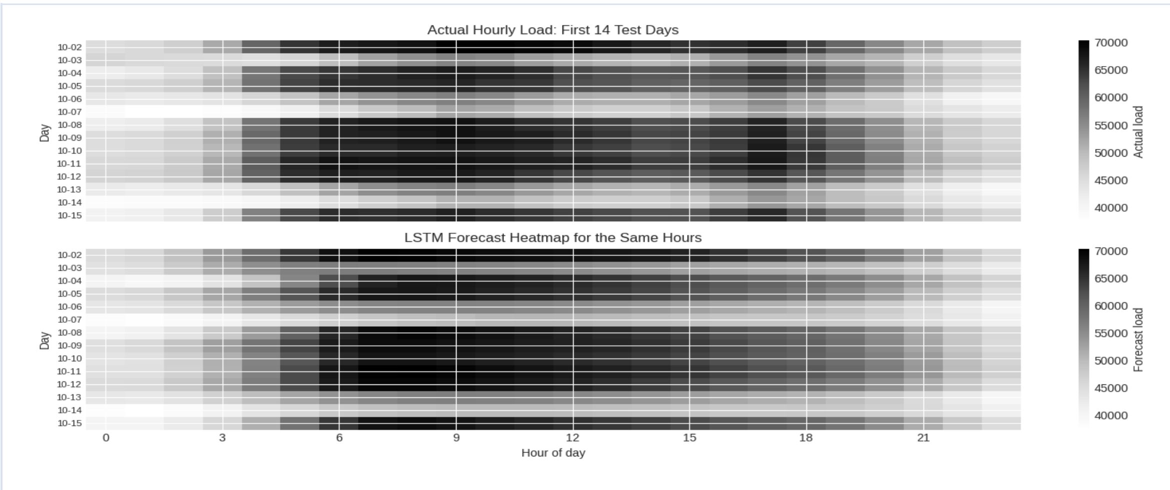


Figure 6. Actual hourly load and LSTM forecast heatmap for the first 14 test days.

5.Results

According to the coding results, the Seasonal Naive benchmark set a strong baseline with RMSE 3006.76, MAE 2318.52 and MAPE 4.41%. Random Forest improved on this benchmark with RMSE 2456.34 and MAPE 3.45%, while Gradient Boosting also improved with RMSE 2484.77 and MAPE 3.58%. RMSE 2117.02 and MAPE 3.02% were attained with the hourly LSTM. After aggregating the hourly LSTM forecasts to weekly averages, the error dropped further to RMSE 328.74 and MAPE 0.48%, although this weekly-aggregated result should be read carefully because averaging smooths hourly errors.

Model	RMSE	MAE	MAPE %
LSTM weekly aggregated	328.74	255.01	0.48
Gradient boosting	2484.77	1874.09	3.58
LSTM	2117.02	1593.38	3.02
Random Forest	2456.34	1799.71	3.45
Seasonal naive	3006.76	2318.52	4.41
SARIMAX full weather + holidays	4908.74	4012.85	7.62
SARIMA	9230.67	7924.20	14.89

Table 3. Forecast accuracy from the current notebook over the final two-year test period.

The rating also conveys a practical message. SARIMA and SARIMAX were good for learning about structure and uncertainty, but the feature-based models were better at using last year's load with minimum adjustments. When accuracy was the sole factor evaluated, the LSTM performed best; nevertheless, compared to a regression model, it needs more preprocessing, calibration, and monitoring.

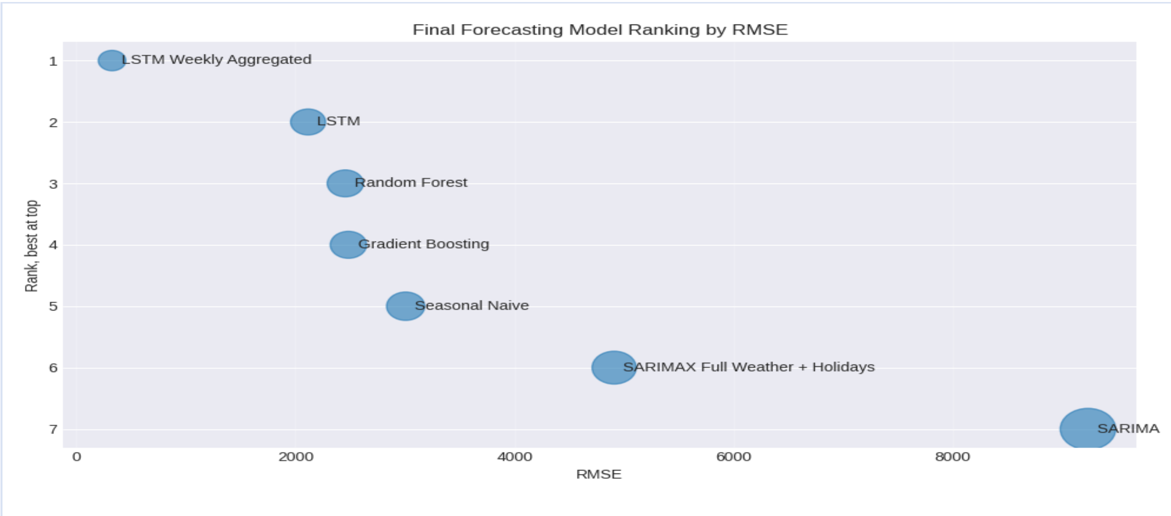


Figure 7. Final forecasting model ranking by RMSE.

The most important lesson is that a more intricate model wasn't always superior. Among the studied requirements, SARIMA had the best AIC; nevertheless, its two-year hold-out accuracy was subpar. This implies that the training sample's likelihood fit did not translate well to the 2018–2020 timeframe. The model may have been overly sensitive to differenced noise because to the high d = 2 order, and 2020 presented unique demand behaviour that was difficult for a model trained primarily on 2015–2018 to predict.

Figure 8. Shape of weekly absolute forecast errors across models.

Although inconsistent, the covariates were helpful. Weather and holidays made SARIMAX far better than normal SARIMA, especially because heating degree days and holiday count had realistic correlations with demand. They were insufficient, though, to surpass seasonal naive. In contrast, the feature models used the same covariates coupled with lags and rolling history, which allowed them to use the yearly memory of the series more efficiently.

However, the weather-based SARIMAX and feature-model results should be regarded as conditional unless actual weather forecasts are utilized in place of the observed weather during the test period. The same-week load from the previous year was the primary predictor, according to the feature importance and coefficients. This makes significant sense: the demand for power is correlated with social habits, employment patterns, and the seasons. The best models were those that maintained this one-year memory while permitting a small amount of temperature and holiday adjustment.

<u>6. Important Analysis:</u>

1. This is due to the German has a seasonal electric electricity. RMSE = 3006.76, MAE = 2318.52, and MAPE = 4.41% were attained. This benchmark was enhanced with Random Forest, Gradient Boosting, and the weekly aggregated LSTM. Gradient Boosting produced RMSE = 2484.77 and MAPE = 3.58%, Random Forest produced RMSE = 2456.34 and MAPE = 3.45%, and the weekly aggregated LSTM produced RMSE = 328.74 and MAPE = 0.48%. The feature-based models and the LSTM produced the significant gains as SARIMA and SARIMAX did not outperform the benchmark.
2. Data leakage was avoided by ensuring that model inputs were derived solely from information that would have been available before the anticipated week. The target week was not inadvertently added as an input since lag load variables were changed prior to modelling. Additionally, rolling averages were solely based on prior weeks. For the LSTM, the scaler was fitted exclusively on the training data and then applied to the test data. Temperature was managed carefully because observed future temperature is not known at the predicted origin. Therefore, unless the temperature values are derived from an actual weather forecast, any SARIMAX result utilizing observed future temperature should be characterized as a conditional or explanatory forecast. Unless future weather forecast values are provided, the weather-based outcomes of the feature models should likewise be considered as conditional because they relied on observed weather variables.
3. The seasonal period was selected at 52 because the main SARIMA/SARIMAX modelling was done on weekly data, and one year has around 52 weekly observations. The EDA plots demonstrated that the demand for electricity in Germany exhibits a distinct yearly pattern, with demand typically being higher during colder months and decreasing during warmer ones. Stationarity was evaluated using the ADF and KPSS tests; the initial weekly series had ADF p = 0.00118 and KPSS p = 0.10. This revealed that the series was not merely a random walk, yet ACF/PACF plots and seasonal decomposition still showed considerable seasonal dependence. Using the necessary AIC grid search over p = 0 to 6, d = 0 to 2, and q = 0 to 6, the final SARIMA structure was chosen, resulting in SARIMA (2,2,6) (1,1,1,52).
4. When compared to ordinary SARIMA, temperature and vacation variables enhanced SARIMAX; nonetheless, this improvement was insufficient to surpass the seasonal naïve benchmark. SARIMA had MAPE = 14.89%, while SARIMAX with full weather and holiday factors reduced this to MAPE = 7.62%. Seasonal naïve, however, continued to outperform with MAPE = 4.41%. This indicates that although the variables included valuable explanatory information, the SARIMAX structure did not make the best use of them to achieve the highest forecast accuracy. Holiday variables are safe for operational forecasting because they are known ahead of time. Temperature is distinct because, unless it is provided by a weather prediction, future measured temperatures are

unknown at the forecast origin. Therefore, unless projected temperature is given, temperature-based SARIMAX should be characterized as conditional.

5. Because its autoaggressive, moving-average, differencing, seasonal, and exogenous components are easily understood, SARIMAX is the most interpretable advanced model. Additionally, it offers confidence intervals, which aid in conveying uncertainty. Moderately interpretable feature-based models are Gradient Boosting and Random Forest Although Random Forest does not offer coefficients in the same manner as a linear regression model, it is simpler to explain than the LSTM because feature importance can be examined. Gradient Boosting is likewise less transparent than SARIMAX, although it still offers feature-importance information.

6. Random Forest is recommended for weekly operational forecasting due to better trade off in terms of accuracy, interpretability and maintenance. It was a significant improvement from the Seasonal Naive benchmark, with RMSE going down and MAPE going down as well about 3006.76 vs 2456.34 on RMSE / 0,044 vs 0.0345 on MAPE. The LSTM had the lowest number of numerical errors, mainly after aggregation into weeks, but incurs additional preprocessing, tuning and monitoring cost. Also, Seasonal Naive, as a simple backup benchmark, should also be retained

7.Conclusion:

This study reveals a strong annual pattern in German electricity demand, making the seasonal naïve model a challenging criterion to surpass. The time-series structure was explained by SARIMA and SARIMAX, and while weather and holiday factors enhanced SARIMAX over simple SARIMA, they were still unable to surpass the benchmark. Random Forest was the best-balanced weekly model since it enhanced forecast accuracy while staying easier to read and maintain than the LSTM. While Random Forest had a somewhat lower RMSE in the final notebook findings, Gradient Boosting also outperformed the Seasonal Naive benchmark.

Overall, the key conclusion is that accurate electricity forecasting requires careful addition of helpful external variables and the capture of annual demand memory. In Future we should include more regional weather data, ongoing validation and probabilistic forecasts to make the model more reliable for real operational planning.

8.References:

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